Sunggyu Ryoo



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Education

Ph.D., Seoul National University — Seoul, Republic of Korea
Department of Physics and Astronomy, Advisor: Takhee Lee — 2024.03–Current

B.S., Seoul National University — Seoul, Republic of Korea
Department of Physics and Astronomy, *Summa Cum Laude* — 2020.03–2024.02

Gyeonggibuk Science High School — Uijeongbu, Republic of Korea
Valedictorian (Early Graduate) — 2018.03–2020.02

Awards and Achievements

- **Outstanding Research Award (Spring 2025, Spring 2026)** — 2025–2026
  - Academic achievements in PhD research
- **SBS Foundation Scholarship** — 2024–Now
  - Merit-based scholarship for PhD students
- **University President's Award (Excellence Award), Seoul National University** — 2024
  - Academic achievements in undergraduate research
- **The Grand Award** — 2023
  - Academic achievements in undergraduate research
- **Dean's List (Spring 2020, Fall 2020, Spring 2021, Fall 2021)** — 2020–2021
  - Academic achievements in undergraduate exams
- **Korean Presidential Scholarship for Science** — 2020–2023
  - The top merit-based scholarship for undergraduate students
- **Hanseong Nobel Scholarship for the gifted** — 2019
  - The top merit-based scholarship for high school students

Research Interests

- 2D Electronic & Optoelectronic Devices
- 2D Doping Techniques

Publications (First author)

1. **Sunggyu Ryoo**, Jongeun Yoo, Seongmin Ko, Jaehyoung Park, Kyungjune Cho*, and Takhee Lee*, "Toward Low-Noise Two-Dimensional Photodetectors: Microscopic Origins and Noise Engineering", in preparation

2. **Sunggyu Ryoo**, Seongmin Ko, Juntae Jang, Donguk Kim, Jaehyoung Park, Yeeun Kim, Youngmin Song, Jaeyong Woo, Jongeun Yoo, Kyungjune Cho*, and Takhee Lee*, "Noise-Suppressed High-Performance WSe_2 Phototransistors enabled by Optically Programmed Remote Contact Doping", *ACS Nano*, in press (2026).

3. Seongmin Ko[||], **Sunggyu Ryoo**[||], Juntae Jang, Jaeyong Woo, Donguk Kim, Jaehyoung Park, Jongeun Yoo, Young Duck Kim*, Kyungjune Cho*, and Takhee Lee*, "Remote Charge Modulation in WSe_2 Transistors via Plasma-Induced Charged Defects in hBN", *ACS Applied Electronic Materials*, in press (2026).

4. Jongeun Yoo[||], Seongmin Ko[||], **Sunggyu Ryoo**[||], Jaehyoung Park, Jaeyoung Kim, Juntae Jang, Jinwoo Sim, Donguk Kim, SeungNam Cha, Chongwu Zhou, Kyungjune Cho*, and Takhee Lee*, "Photoresponse Properties of Ambipolar Transport in WSe_2 Field-Effect Transistors", *Advanced Electronic Materials*, 12, e00669 (2026).

5. **Sunggyu Ryoo**[||], Taehee Kim[||], Jongeun Yoo, Seongmin Ko, Kyungjune Cho*, and Takhee Lee*, "Enhancing Photodetection of WSe_2 Phototransistors by Reducing Charge Scattering with hBN Passivation", *Nano Futures*, 9, 035001 (2025).

6. **Sunggyu Ryoo**, Jinwoo Sim, Seungjun Jeong, Juntae Jang, Jaeyong Woo, Jaehyoung Park, Seongmin Ko, Yeeun Kim, Youngmin Song, Jongeun Yoo, Heebeom Ahn, Keehoon Kang, Daeheum Cho, Kyungjune Cho*, and Takhee Lee*, "Noise-Reduced WSe_2 Phototransistors for Enhanced Photodetection Performance via Suppression of Metal-Induced Gap States", *Advanced Materials Technologies*, 10, 2500064 (2025).

7. Jinwoo Sim[||], **Sunggyu Ryoo**[||], Joo Sung Kim[||], Juntae Jang, Heebeom Ahn, Donguk Kim, Joonha Jung, Taehyun Kong, Hyeonmin Choi, Yun Seog Lee, Tae-Woo Lee*, Kyungjune Cho*, Keehoon Kang*, and Takhee Lee*, "Enhanced Photodetection Performance of In Situ Core/Shell Perovskite-MoS_2 Phototransistor", *ACS Nano*, 18, 16905 (2024).

Publications (Co-author)

1. Jaehyoung Park, Juntae Jang, Seongmin Ko, **Sunggyu Ryoo**, Jongeun Yoo, Kyungjune Cho*, and Takhee Lee*, "Universal Ohmic Contacts in Electrochemically Exfoliated MoS_2 Enabled by Defect-Induced Gap States", in preparation.

2. Jongeun Yoo, Jaeyoung Kim, Jinwoo Sim, Jaehyoung Park, Seongmin Ko, **Sunggyu Ryoo**, Jae-Keun Kim*, Kyungjune Cho*, and Takhee Lee*, "Avalanche Multiplication in 2D Materials and Its Device Applications: From Scattering-Limited to Ballistic Transport Regimes", in preparation.

3. Yeeun Kim[||], Youngmin Song[||], Hyemin Lee, **Sunggyu Ryoo**, Jaeyong Woo, Youcheng Zhang, Henning Sirringhaus, Keehoon Kang*, and Takhee Lee*, "Defect-Induced Universal Hole Injection in 2D Tin Halide Perovskite Field-Effect Transistors", *ACS Applied Materials & Interfaces*, in revision.

4. Youngmin Song, Jaeyong Woo, Hyemin Lee, Jaehyoung Park, Seongmin Ko, Yeeun Kim, **Sunggyu Ryoo**, Henning Sirringhaus, Keehoon Kang*, and Takhee Lee*, "Impact of Gate-Induced Mobility Attenuation on Accurate Characterization of Perovskite Field Effect Transistors", *ACS Applied Materials & Interfaces*, in press (2026).

5. Jaeyong Woo, Seongmin Ko, Youngmin Song, Yeeun Kim, **Sunggyu Ryoo**, Taehyun Kong, Keehoon Kang*, and Takhee Lee*, "Dopant- and Trap-Limited Gate Modulation in 2D Tin Halide Perovskite Field-Effect Transistors", *ACS Applied Electronic Materials*, 8, 6156 (2026).

6. Hyeonmin Choi[||], Joonha Jung[||], Young-Kwang Jung[||], Seok Woo Lee, Taehyun Kong, Yongjin Kim, Youjin Reo, Jinwoo Sim, Yeeun Kim, Jaeyong Woo, **Sunggyu Ryoo**, Jaeyoon Cho, Youcheng Zhang, Stefano Pecorario, Bo Ram Lee, Henning Sirringhaus, Yong-Young Noh, Samuel Stranks, Takhee Lee*, and Keehoon Kang*, "Strong-Binding Small-Molecule Passivator for Two-Dimensional Tin-Based Perovskite Field-Effect Transistors", *Adv Func Mater*, 36, e74094 (2026).

Presentations

1. **Sunggyu Ryoo**, Seongmin Ko, Juntae Jang, Donguk Kim, Jaehyoung Park, Youngmin Song, Yeeun Kim, Jaeyong Woo, Jongeun Yoo, Kyungjune Cho, and Takhee Lee, "Optical Contact Doping for Low-Noise WSe_2 Phototransistors", The 34th Korean Conference on Semiconductors (KCS 2027), Gangneung Convention Center, Gangneung, Korea, January 26-29, 2027. [oral talk]

2. **Sunggyu Ryoo**, Seongmin Ko, Juntae Jang, Donguk Kim, Jaehyoung Park, Youngmin Song, Yeeun Kim, Jaeyong Woo, Jongeun Yoo, Kyungjune Cho, and Takhee Lee, "Photo-Induced Contact Doping for Low-Noise and High-Performance WSe_2 Phototransistors", 2026 MRS (Materials Research Society) Fall Meeting, Hynes Convention Center, Boston, USA, November 29-December 4, 2026. [oral talk]

3. **Sunggyu Ryoo**, Seongmin Ko, Juntae Jang, Donguk Kim, Jaehyoung Park, Youngmin Song, Yeeun Kim, Jaeyong Woo, Jongeun Yoo, Kyungjune Cho, and Takhee Lee, "High-Performance WSe_2 Phototransistors with Optically Programmed Remote Contact Doping", 2026 KPS (Korean Physical Society) Fall Meeting, Jeju International Convention Center, Jeju, Korea, October 20-23, 2026. [oral talk]

4. **Sunggyu Ryoo**, Jinwoo Sim, Taehee Kim, Juntae Jang, Seungjun Jeong, Daeheum Cho, Kyungjune Cho, and Takhee Lee, "Noise Reduction in WSe_2 Phototransistors for Enhanced Photodetection", The 33rd Korean Conference on Semiconductors (KCS 2026), High One resort, Jeongseon, Korea, January 27-30, 2026.

5. **Sunggyu Ryoo**, Jinwoo Sim, Joo Sung Kim, Tae-Woo Lee, Kyungjune Cho, Keehoon Kang, and Takhee Lee, "High-Performance Phototransistor based on In-Situ Core/Shell Perovskite-MoS_2 Heterostructure", The 15th International Conference on Flexible and Printed Electronics (ICFPE 2025), Institute of Science Tokyo, Japan, September 17-19, 2025. [Best Poster Award]

6. **Sunggyu Ryoo**, Jinwoo Sim, Joo Sung Kim, Tae-Woo Lee, Kyungjune Cho, Keehoon Kang, and Takhee Lee, "High-Performance MoS_2 Phototransistor using In-Situ Core/Shell Perovskite with Photogating Effect", International Symposium on Organic and Perovskite Electronics (ISOPE 2025), Seoul National University, Seoul, Korea, July 30-August 1, 2025.

7. **Sunggyu Ryoo**, Jinwoo Sim, Seungjun Jeong, Juntae Jang, Keehoon Kang, Daeheum Cho, Kyungjune Cho, and Takhee Lee, "Performance-Enhanced WSe_2 Phototransistors through Noise Reduction using Semimetal Contacts", 2025 KPS (Korean Physical Society) Spring Meeting, Daejeon Convention Center, Daejeon, Korea, April 23-25, 2025. [oral talk]

8. **Sunggyu Ryoo**, Jinwoo Sim, Joo Sung Kim, Juntae Jang, Heebeom Ahn, Tae-Woo Lee, Keehoon Kang, Kyungjune Cho, and Takhee Lee, "High-Performance Photodetection of In-Situ Core/Shell Perovskite-MoS_2 Phototransistors", APS (American Physical Society) Global Physics Summit 2025 March Meeting, Anaheim Convention Center, Anaheim, USA, March 16-21, 2025. [oral talk]

9. **Sunggyu Ryoo**, Jinwoo Sim, Jaeyong Woo, Jaehyoung Park, Yeeun Kim, Youngmin Song, Heebeom Ahn, Kyungjune Cho, and Takhee Lee, "Semimetal Contact Engineering for Low-Noise, High-Performance WSe_2-Based Phototransistors", The 32nd Korean Conference on Semiconductors (KCS 2025), High One resort, Jeongseon, Korea, February 12-14, 2025. [oral talk]

10. **Sunggyu Ryoo**, Jinwoo Sim, Joo Sung Kim, Juntae Jang, Tae-woo Lee, and Takhee Lee, "Design and Fabrication of an In-situ Core/Shell Perovskite/MoS_2 Heterostructure for High-Performance Photodetection", The 31st Korean Conference on Semiconductors (KCS 2024), Gyeongju Hwabaek International Convention Center, Gyeongju, Korea, January 24-26, 2024. [oral talk]

11. **Sunggyu Ryoo**, Jinwoo Sim, Joo Sung Kim, Juntae Jang, Tae-Woo Lee, and Takhee Lee, "Enhancing Photodetection Performance with In-Situ Core/Shell Perovskite-MoS_2 Heterostructure", NANO KOREA 2023, KINTEX, Ilsan, Korea, July 5-7, 2023.

Teaching

- **Teaching Assistant** at Seoul National University (Excellent TA Award) — Fall 2025
  College Physics 2
- **Teaching Assistant** at Seoul National University — Spring 2025
  College Physics 1
- **Teaching Assistant** at Seoul National University (Excellent TA Award) — Fall 2024
  Physics Lab.2
- **Teaching Assistant** at Seoul National University — Spring 2024

Physics Lab.1

- **Tutor** at Seoul National University Fall 2021
  College Writing 2: Writing in Science & Technology